University of Glasgow
School of Computing Science

Blockchain-Based Electronic Voting

Ten Key Figures

A guided visual summary of the dissertation's results,
with each figure explained in plain language

MSc Computing Science

Dissertation — visual summary

The dissertation's ten figures, each explained in plain language: what the model is, how it was checked, what it found, and what a defender should do about it. Numbering matches the dissertation exactly, and every number quoted is injected from the simulation output (seed 42, 2,500 paired trials).

How to read this document

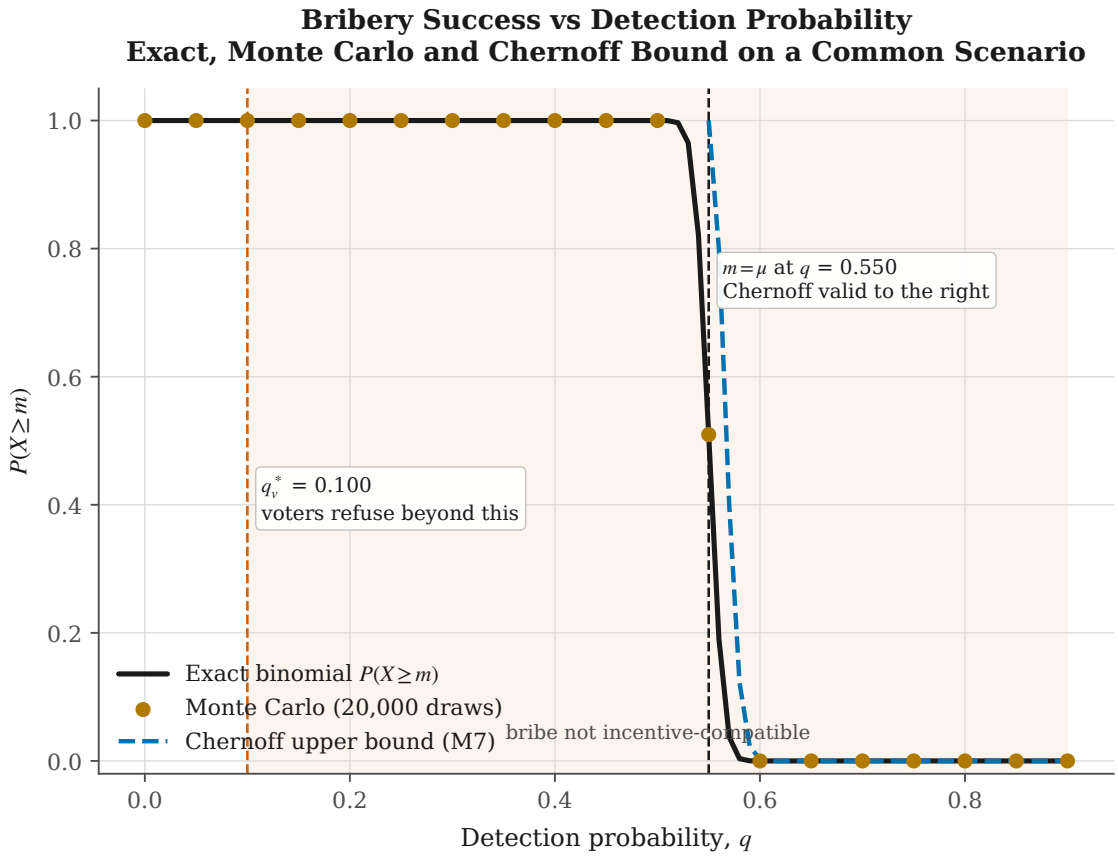
Each of the following ten pages carries one figure and three short notes: how to read it, what it shows, and why it matters. A fourth note gives the single number worth remembering from that figure.

Section numbers here are the dissertation's own figure numbers, so Section 7 of this guide explains Figure 7 of the dissertation. The dissertation uses these ten figures and no others.

THE CONVENTIONS, ONCE Vermillion is always the **baseline** (weakly defended) system and blue is always the **protected** one. Charcoal dashed lines are always **thresholds**, and each is labelled with its numeric value. The three attack vectors keep their own colour and marker throughout: bribery vermillion circles, Sybil green squares, ballot stuffing purple triangles. The palette was checked programmatically for colour-blind separation, and every figure is also readable in greyscale because colour is never the only channel.

Fig.	Subject	Answers
1	Three ways to compute the same number	Can it be trusted?
2	Why bribery collapses: the voter's decision	What drives the result?
3	Both Chernoff directions validated	Is the bound sound?
4	What the countermeasures actually do	RQ3
5	From probabilities to stolen elections	Does it happen for real?
6	How much detection is actually needed	RQ4
7	The model in one picture	What is being computed?
8	The rational attacker does not spend it all	RQ1
9	Why no single vector is enough	Setup for RQ5
10	The coalition attack	RQ5

1. Three ways to compute the same number



Fixed scenario: N = 2,000, m = 900, B = 200,000, R = 100, P = 900, giving T = 2,000 attempts. Only q varies.

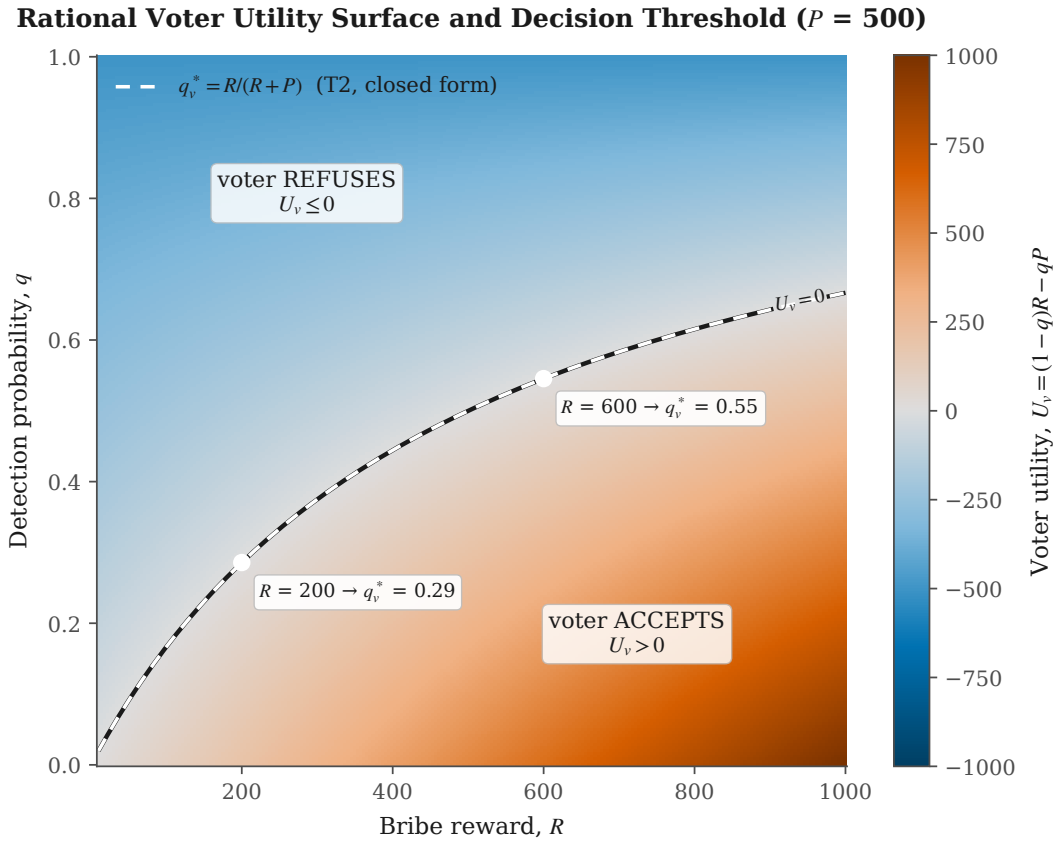
WHAT YOU ARE LOOKING AT One fixed election scenario, with only the detection probability q varying along the horizontal axis. Three methods compute the same success probability: the exact binomial calculation (solid black), a Monte Carlo simulation (gold markers), and the Chernoff upper bound (dashed blue). The shaded region to the right is where the bribe is no longer worth accepting.

WHAT IT SHOWS The Monte Carlo estimate is indistinguishable from the exact curve, and the bound sits above both wherever it is defined — valid, but visibly conservative. It also shows the shape of the detection effect: success stays near certainty and then *collapses* over a narrow band of q, rather than declining steadily.

WHY IT MATTERS Two things at once. Methodologically, three independent computations agreeing is strong evidence that none contains an error. Substantively, the collapse is why detection is the efficient defensive lever: a small improvement in the right region does far more than a large improvement anywhere else.

THE NUMBER TO REMEMBER Mean absolute Monte Carlo error against the exact value: 7.98e-5 for bribery. The simulation and the mathematics agree to four decimal places.

2. Why bribery collapses: the voter's decision



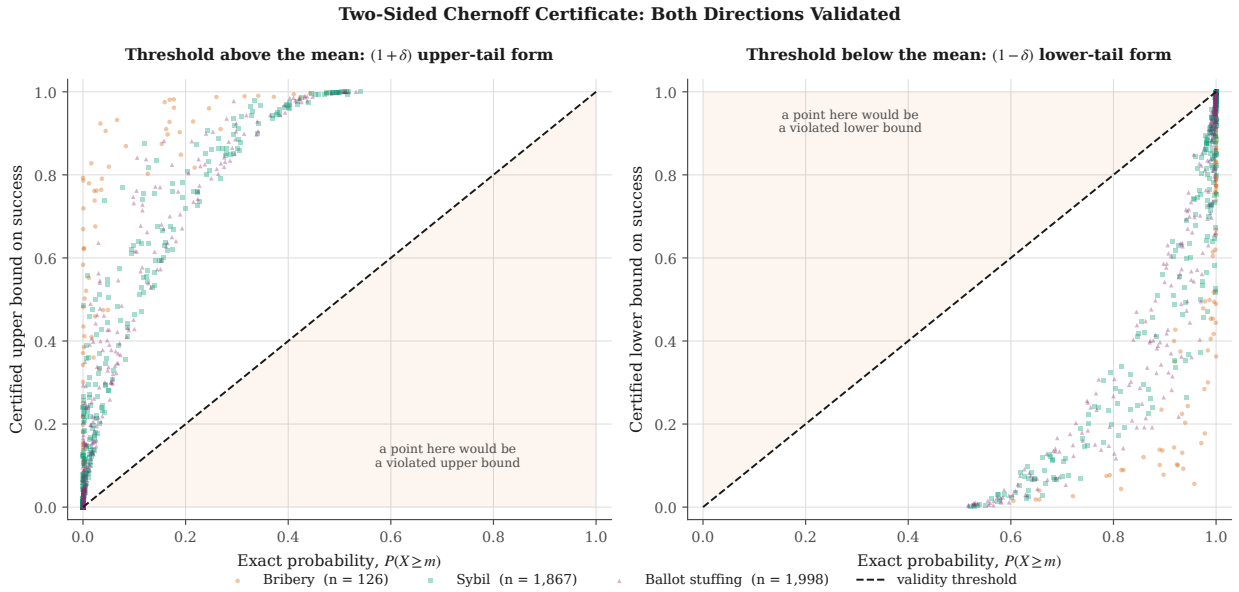
WHAT YOU ARE LOOKING AT A map of one voter's decision. Left to right is the size of the bribe offered; bottom to top is the probability of being caught. Warm colours mean accepting the bribe pays; cool colours mean it does not. The black curve is the break-even line $U_v = 0$, and the white dashed curve is the closed-form threshold $q_v^* = R/(R+P)$ drawn on top of it.

WHAT IT SHOWS Bribery has a consent gate that the other two attacks do not. Above the threshold curve, a rational voter refuses *any* bribe the attacker can afford, so the attack stops — not because fraud was detected, but because it was never committed. That the numerically computed boundary and the algebraic formula land on the same curve is also a check that the solver and the mathematics agree.

WHY IT MATTERS It explains why bribery behaves discontinuously in every later figure while Sybil and ballot-stuffing attacks degrade smoothly. Bribery is suppressed twice — once by detection, once by refusal — which is why it ends up the *least* dangerous of the three vectors under protection, despite being the one most often discussed.

THE NUMBER TO REMEMBER $q_v^* = R/(R+P)$ depends only on the ratio of bribe to penalty — not on the electorate size, the margin, or the attacker's budget. Raise the penalty and the whole acceptance region shrinks.

3. Both Chernoff directions validated



Every one of the 15,000 scenario evaluations receives a certified statement, and the certified interval contains the exact value in 15,000 of 15,000 cases. The one-sided bound alone covers only the left panel.

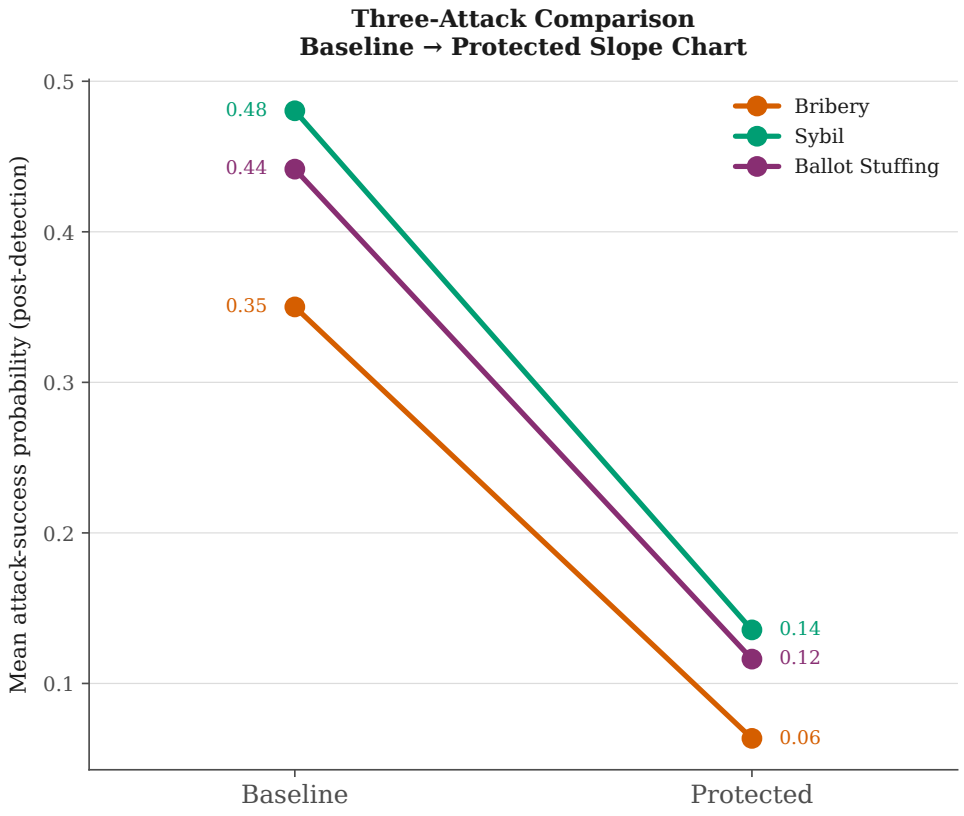
WHAT YOU ARE LOOKING AT Every scenario in the experiment plotted twice over: its exact success probability horizontally, and the guarantee proved about it vertically. Left, where the threshold sits *above* the mean, the $(1+\delta)$ upper-tail bound certifies success from above, so valid points lie on or above the diagonal. Right, where the threshold sits *below* the mean, the $(1-\delta)$ lower-tail form certifies success from below, so valid points lie on or below it.

WHAT IT SHOWS Both directions hold everywhere. No point falls on the wrong side of either diagonal. The two forms are not interchangeable — the factor records which side of the mean the threshold is on — and applying either to the wrong tail produces a statement the data refutes.

WHY IT MATTERS The one-sided bound alone covers only the left panel, leaving the scenarios where the attack is *most likely to succeed* with no proof attached to them at all. Extending to both tails means every scenario now carries a guarantee, and in the right-hand panel that guarantee is the stronger kind: not "the attack cannot exceed this chance" but "the attack succeeds with *at least* this chance".

THE NUMBER TO REMEMBER 100% — the certified interval contains the exact value in all 15,000 scenario evaluations, and none is left without a statement.

4. What the countermeasures actually do



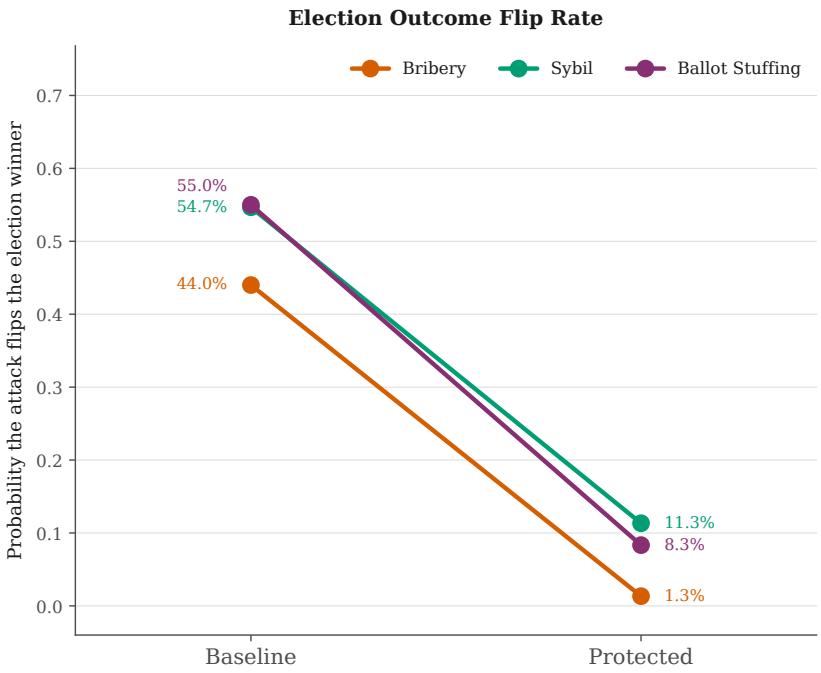
WHAT YOU ARE LOOKING AT Each line is one attack vector. The left end is its average success probability against the weakly-defended baseline system; the right end is the same scenario re-run with blockchain-style audit, digital signatures, identity verification and ballot validation switched on. Both ends of every line come from the *same* 2,500 scenarios, so the drop is a like-for-like comparison.

WHAT IT SHOWS All three attacks fall sharply — but not equally. Bribery falls furthest in relative terms, because it is stopped twice over. Sybil registration ends up the most resilient vector, retaining roughly twice the residual success probability of bribery.

WHY IT MATTERS The ordering is the finding, not the size of the drop. The weakest link under protection is *identity verification* — the one thing a ledger does nothing about. A deployment that hardens the chain while leaving voter registration weak has hardened the wrong layer.

THE NUMBER TO REMEMBER Bribery 0.350 → 0.064 (81.9%), Sybil 0.480 → 0.136 (71.8%), ballot stuffing 0.442 → 0.116 (73.7%).

5. From probabilities to stolen elections



Measured on genuinely mined chains with real tallies and real winner determination, not on abstract success probabilities.

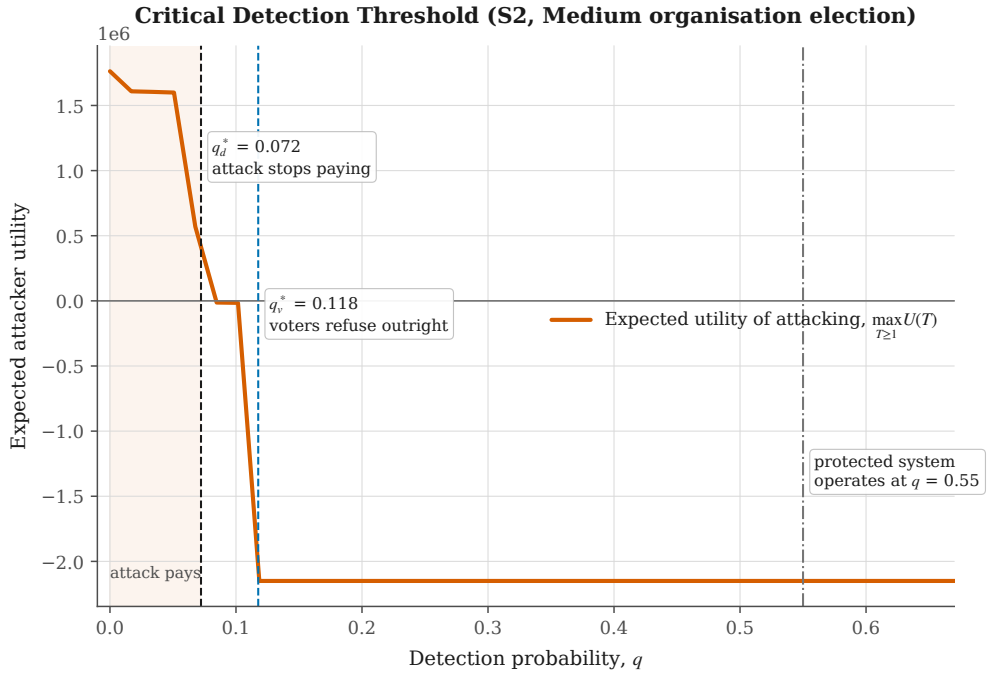
WHAT YOU ARE LOOKING AT Not a probability model this time. 300 elections were genuinely mined on a proof-of-work blockchain, attacked, and re-counted from the chain itself to see who actually won. The vertical axis is the fraction of those elections in which the *winner changed*.

WHAT IT SHOWS The ordering matches the abstract results exactly, which is the cross-check the two-layer design exists to provide: an analytic model and a mined chain, computed independently, agree. Under the baseline configuration between two and three elections in five were flipped.

WHY IT MATTERS This is the centrepiece of the dissertation, and it is a negative result about the technology. The chain's integrity check passed in *every single trial*, including all the ones where the winner changed. Every fraudulent vote was correctly formed, correctly mined and permanently recorded. The ledger did exactly what a ledger is supposed to do, and the elections were stolen anyway.

THE NUMBER TO REMEMBER 44.0% of baseline bribery trials flipped the winner while chain integrity was never violated. Immutability guarantees a wrong result cannot later be altered — not that the result is right.

6. How much detection is actually needed



Only about 7.2% detection is needed to make this attack unprofitable, well below the 11.8% at which voters would refuse outright, and far below the 55% the protected configuration actually achieves.

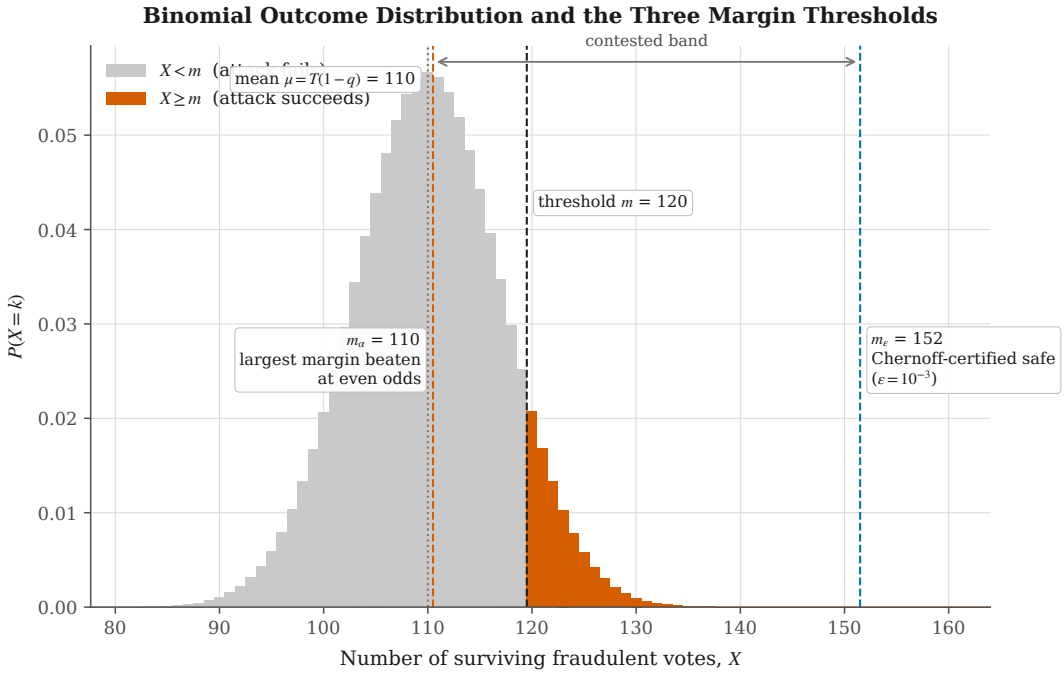
WHAT YOU ARE LOOKING AT The attacker's expected profit as the system's detection probability improves left to right. The shaded region is where attacking pays. Three vertical markers: q_d^*, where profit hits zero and the attack stops being worth mounting; q_v^*, where voters refuse outright and it becomes impossible; and the detection rate the fully protected system actually delivers.

WHAT IT SHOWS The two thresholds are far apart, and both are far to the left of what the protected system provides. Between q_d^* and q_v^* the attack is still perfectly *possible* — the budget suffices, the voters would accept — but a rational attacker declines because the expected return has gone negative.

WHY IT MATTERS This is the most actionable finding in the dissertation, and it inverts the usual conclusion. The headline result is not that the countermeasures work; it is that *much weaker ones would have*. Deterrence arrives long before prevention, and costs far less. The caveat is that deterrence assumes a rational attacker: someone with non-monetary motives, or who misjudges the detection rate, is not deterred by an unfavourable expectation.

THE NUMBER TO REMEMBER Attacks stop paying at detection probabilities of 0.050–0.118 — roughly 4.7× weaker than the 0.55 the protected system delivers. About one caught bribe in fourteen is enough.

7. The model in one picture



$X \sim \text{Binomial}(T = 200, 1 - q = 0.55)$; the shaded success region has probability $P(X \geq m) = 0.088$. Because X is integer-valued, region boundaries are drawn at half-integers (the m line sits at $m - 0.5$), so each line meets the edge of a bar rather than cutting through its centre. Margins at or below m_α are comfortably within reach; margins at or above m_ε are provably out of reach.

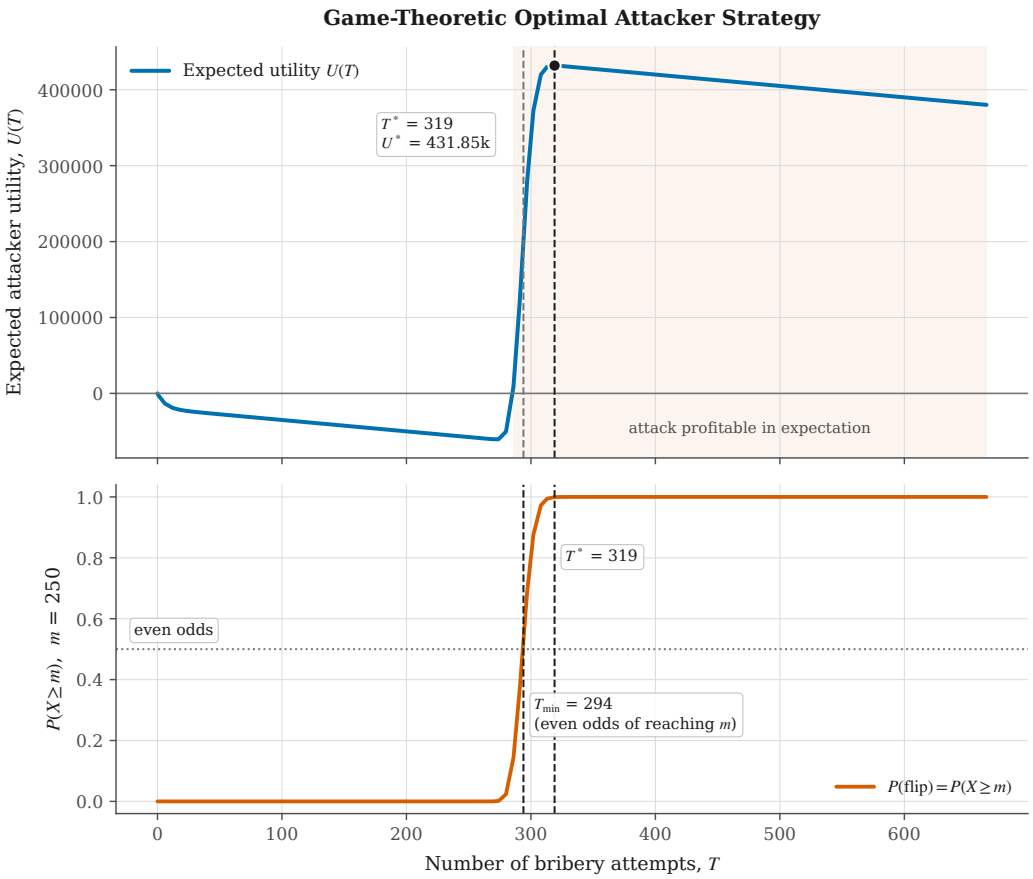
WHAT YOU ARE LOOKING AT The bell-shaped curve is the probability of each possible number of fraudulent votes surviving the system's defences. The grey bars are outcomes where the attack *fails*; the vermillion bars are outcomes where it *succeeds*, because they reach or exceed the manipulation threshold m — the number of extra votes needed to overturn the election. The shaded vermillion area is the attack's success probability.

WHAT IT SHOWS Attack success is a tail event, not an average. The mean outcome μ sits comfortably below the threshold, so a typical attack fails; the attack succeeds only when the random outcome runs unusually high. The three labelled thresholds partition the axis: below m_α the attacker wins at even odds, above m_ε the Chernoff bound proves safety, and only the contested band between them needs exact computation.

WHY IT MATTERS This one picture defines every number in the dissertation. "Attack success" always means reaching m — never merely that some fraud occurred, or that it went undetected. It is a deliberately demanding definition: an attack that injects a hundred fraudulent votes into an election it fails to flip counts here as a *failure*.

THE NUMBER TO REMEMBER Margins are safe or contested by a hair. At national scale the attacker reaches 19,000 votes at even odds while safety is only provable from 19,515 — a contested band about 500 votes wide.

8. The rational attacker does not spend everything



Fixed scenario: $N = 2{,}000$, $m = 250$, $B = 100{,}000$, $R = 150$, $q = 0.15$. Utility peaks at $T^* = 319$ and then declines: once the flip probability saturates, each further bribe adds cost and detection risk but no additional chance of success.

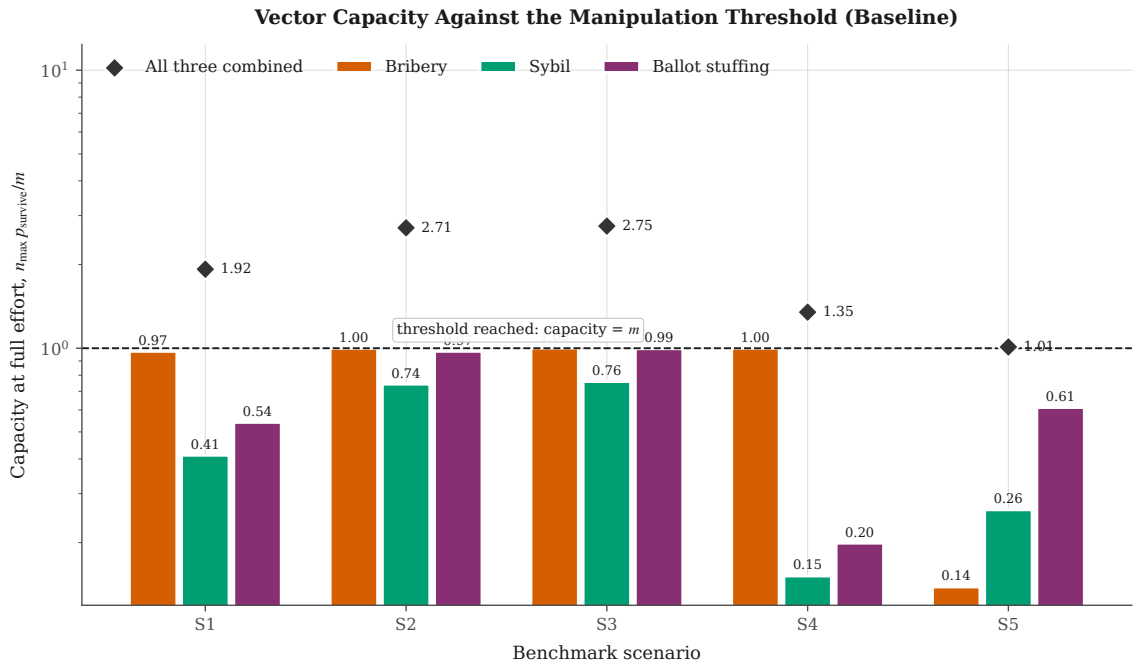
WHAT YOU ARE LOOKING AT Two stacked panels sharing a horizontal axis: the number of bribes the attacker attempts. The top panel is the attacker's expected profit, the bottom panel is the probability of flipping the election. T^* marks the most profitable strategy, and T_{min} the effort needed for an even chance of success.

WHAT IT SHOWS Profit is single-peaked. It rises while extra bribes still move the flip probability, then *falls* once that probability saturates near one and every further bribe adds cost and detection risk without adding any chance of success. Spending the whole budget is a strictly worse strategy than stopping at T^*.

WHY IT MATTERS Security analyses usually assume a maximally-resourced attacker who spends everything. That attacker is not the dangerous one — the dangerous one stops early, spends less, and is harder to notice. Optimising a defence against maximum expenditure optimises against the wrong opponent.

THE NUMBER TO REMEMBER Utility peaks well before the budget is exhausted, and declines thereafter. The defence question is not "can they afford it?" but "is it worth their while?"

9. Why no single vector is enough



Values above 1.0 mean the vector can reach the threshold on its own. Sybil and ballot stuffing fall below the line in every scenario once their registration and turnout ceilings are respected, yet their combined capacity clears it -- no single vector suffices, the mixture does.

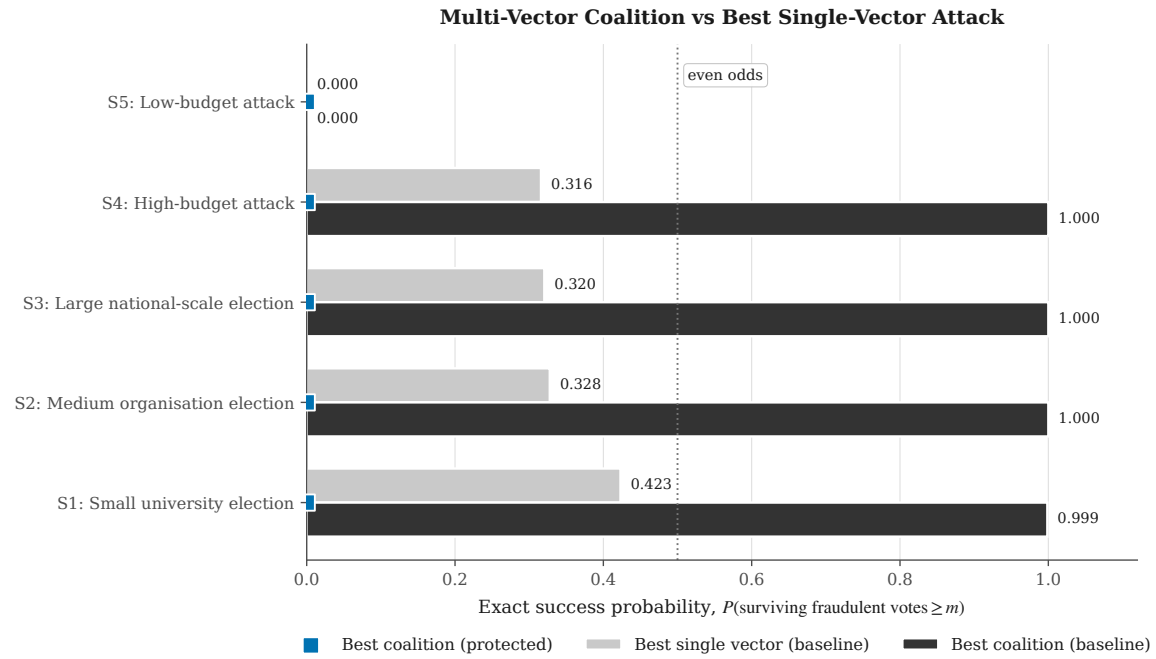
WHAT YOU ARE LOOKING AT For each benchmark scenario, how many fraudulent votes each vector could deliver at maximum effort, expressed as a multiple of the threshold needed to flip the election. The dashed line at 1.0 is that threshold: a bar reaching it means the vector can win on its own. The diamond shows all three vectors' capacity added together.

WHAT IT SHOWS Neither Sybil registration nor ballot stuffing reaches the line in *any* scenario, once realistic limits are respected — forged registrations bounded by the registration window, injected ballots bounded by the turnout figure they would otherwise make implausible. Individually, two of the three attacks are structurally infeasible. The combined diamond clears the line comfortably.

WHY IT MATTERS Read one vector at a time, this is a reassuring result: most of the attacks cannot work. That reading is wrong, and the next page shows why. The limits binding each vector are *capacity* limits, and capacity limits do not add up — an attacker who spreads effort escapes all three at once.

THE NUMBER TO REMEMBER Sybil and ballot stuffing individually reach only 0.14–0.99 of the threshold. Together with bribery they reach 1.01–2.75.

10. The coalition attack



Protected coalition markers sit at zero in every scenario: the countermeasure configuration suppresses the mixed attack as completely as it suppresses each single vector.

WHAT YOU ARE LOOKING AT For each scenario, two bars and a marker. The grey bar is the best the attacker can do committing the whole budget to a single vector; the dark bar is the best achievable by splitting that same budget across all three at once. The blue square is the mixed attack re-run against the protected system.

WHAT IT SHOWS Mixing wins, and not by a little. In the high-budget scenario no single vector exceeds a success probability of 0.316, yet the mixture reaches near-certainty. The mechanism is capacity, not probability: each vector saturates its own ceiling long before the money runs out, and the leftover budget buys nothing more there but a great deal elsewhere.

WHY IT MATTERS Evaluating attack vectors one at a time is *not* the conservative choice it appears to be. Doing so here would have declared the attack infeasible when in fact it succeeds decisively. The reassuring news is that the same countermeasures still hold: every protected marker sits at zero.

THE NUMBER TO REMEMBER The mixed strategy beats the best single vector in 4 of 5 scenarios, lifting mean success from 0.277 to 0.800 — and the result survives all 108 cost configurations tested.

The argument in six sentences

1. An attack succeeds only if enough fraudulent votes survive every defence to overturn the margin, which makes election security a question about a probability distribution rather than a binary vulnerability.
2. Bribery is stopped by voters refusing as much as by detection, which is why it ends up the least dangerous of the three vectors once defences are switched on.
3. Three independent computational methods — exact, simulated and proved — agree everywhere they are all defined, and the proved bound now covers both tails, so every scenario carries a guarantee.
4. The modelled countermeasures cut attack success by 71.8–81.9%, but the residual risk concentrates in identity verification, which is precisely what a ledger does not address.
5. Chain integrity held in every one of 300 mined elections, and 44.0% of the baseline ones were stolen regardless — immutability and correctness are different properties.
6. Making an attack unprofitable takes roughly 4.7× less detection than making it impossible, and an attacker who combines vectors beats one who commits to the best single vector in 4 of 5 scenarios — so defenders should buy deterrence first, and should never evaluate attack vectors in isolation.

The full dissertation, including the literature review, the formal model, the methodology and all thirty figures, is available separately. Every number in this summary is injected at build time from the simulation output and is reproducible from seed 42.